

Optimization Techniques in Power System (BT EE610)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

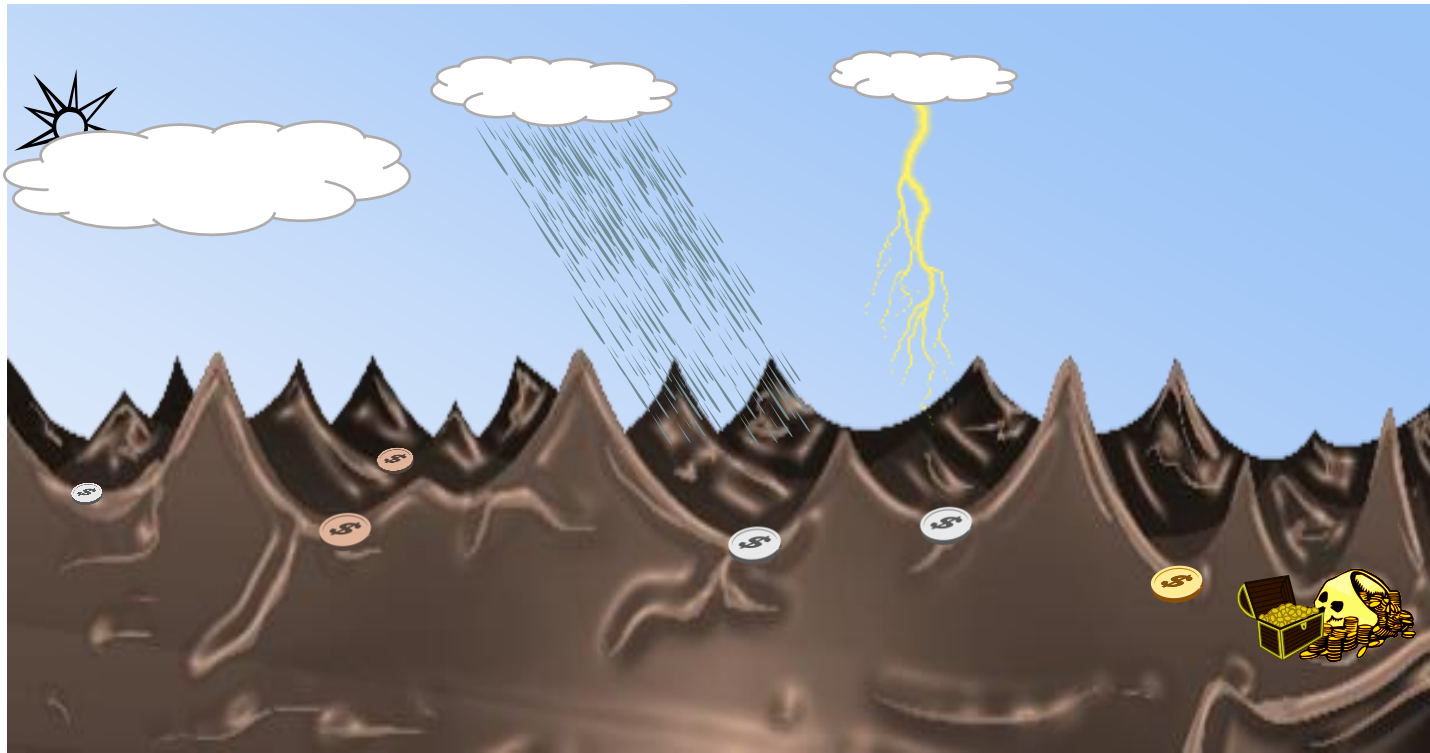
Unit-II

Single-objective optimization algorithms

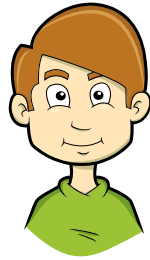
By
Dr. Manish Kumar
Asth. Professor
Department of Electrical Engineering
SOE&T, CUH

**Single-objective
optimization
algorithms**

Analogy



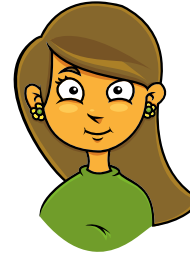
Analogy



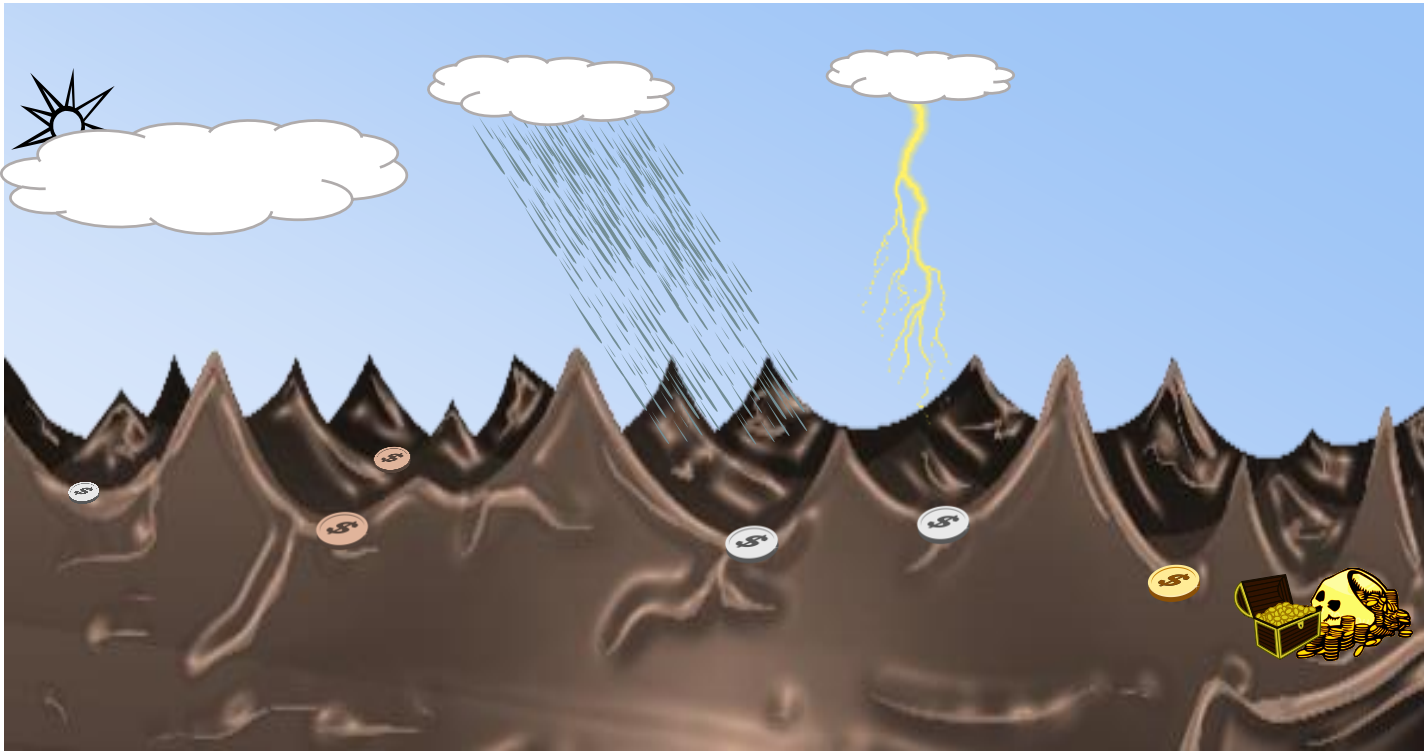
BAJ team: Bob



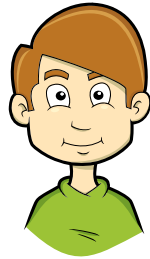
Anthony



Jennifer



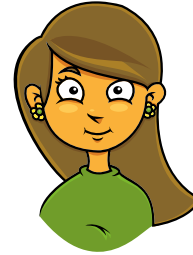
Analogy



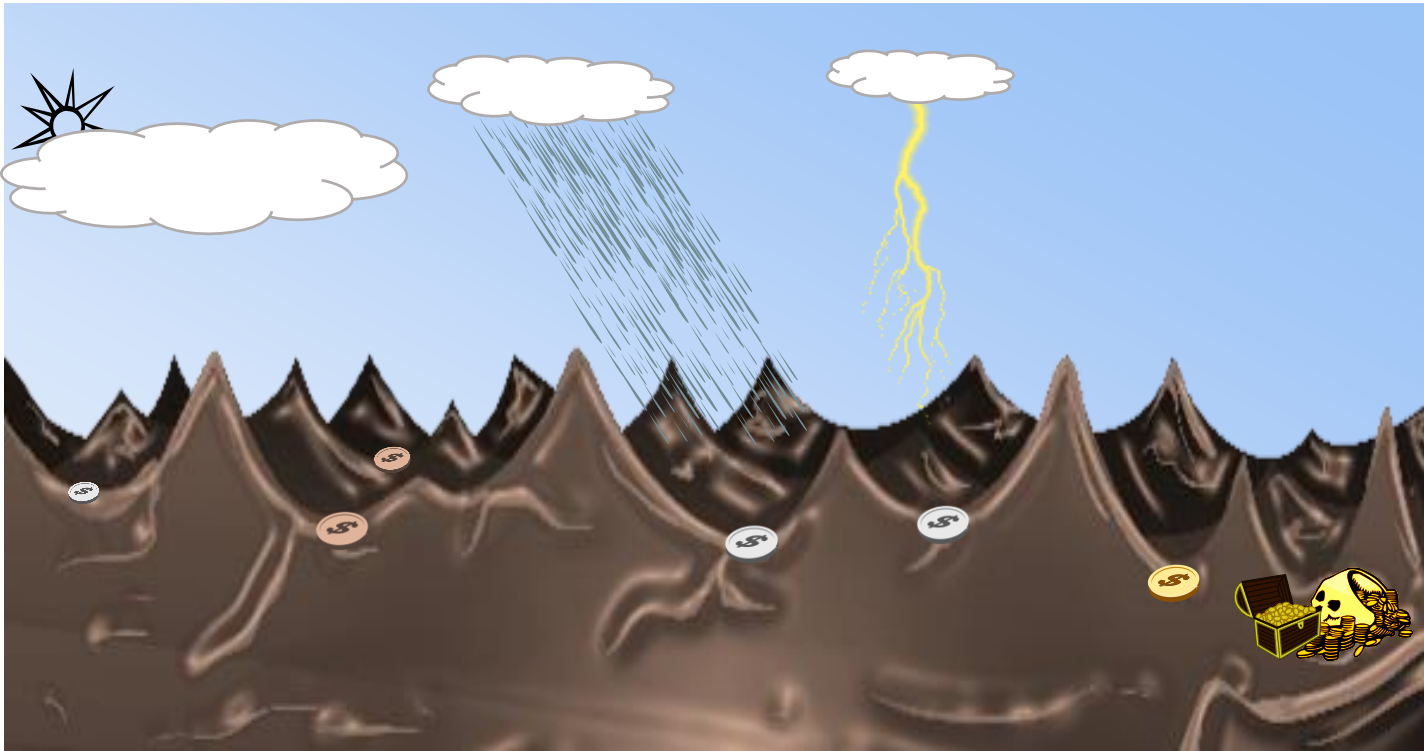
BAJ team: Bob



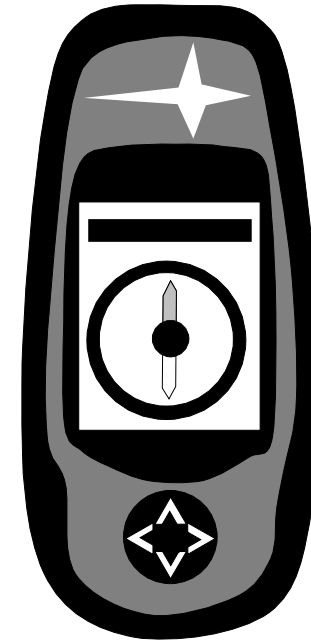
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Jennifer



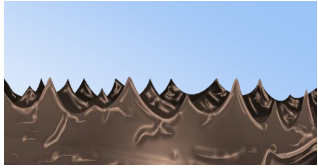
GPS



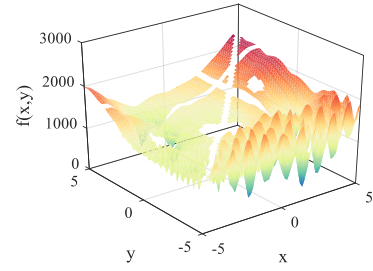
Latitude (x)
Longitude (y)
Elevation (z)

Analogy

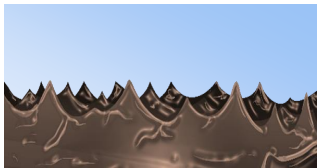
Analogy



Landscape



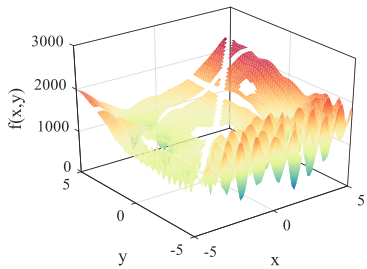
Analogy



Landscape

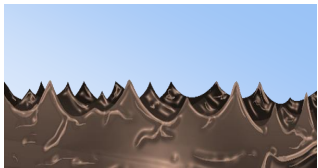


GPS: x,y



Variables

Analogy



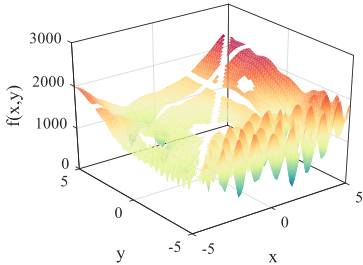
Landscape



GPS: x,y



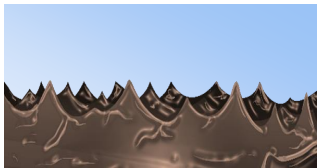
GPS: z



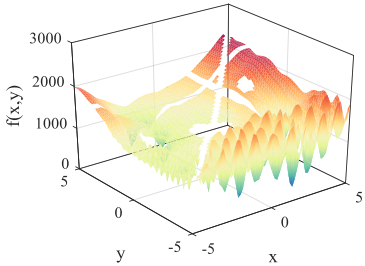
Variables

Objective value

Analogy



Landscape



Variables



GPS: x,y



GPS: z

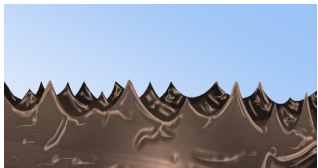
Objective value



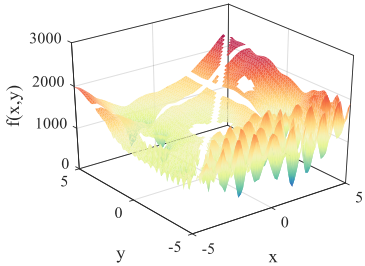
GPS

Objective function

Analogy



Landscape



Variables



GPS: x,y



GPS: z

Objective value



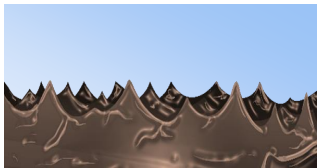
GPS

Objective function

All the possible
locations (x,y)

Search space

Analogy



Landscape



GPS: x,y

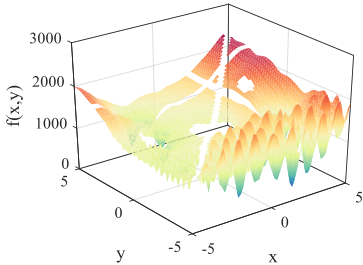


GPS: z



GPS

All the possible
locations (x,y)



Variables

Objective value

Objective function

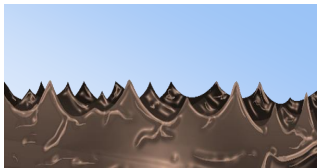
Search space



Rain, flood, etc.

Constraints

Analogy



Landscape



GPS: x,y

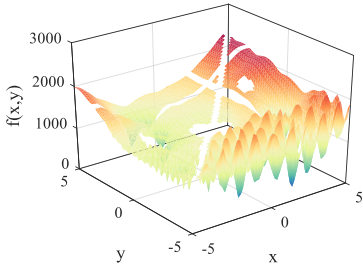


GPS: z



GPS

All the possible
locations (x,y)



Variables

Objective value

Objective function

Search space



Rain, flood, etc.

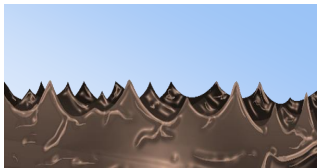
Constraints



Treasure

Global optimum

Analogy



Landscape



GPS: x,y

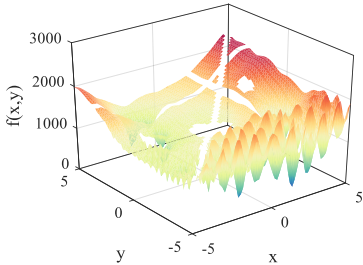


GPS: z



GPS

All the possible
locations (x,y)



Variables

Objective value

Objective function

Search space



Rain, flood, etc.

Constraints



Treasure

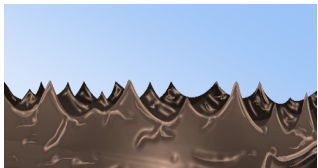
Global optimum



Small coins

Local optima

Analogy



Landscape



GPS: x,y

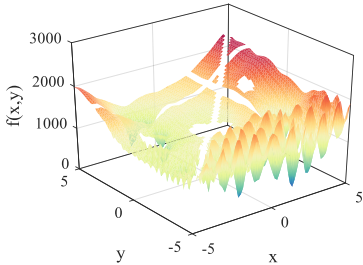


GPS: z



GPS

All the possible
locations (x,y)



Variables

Objective value

Objective function

Search space



Rain, flood, etc.

Constraints



Treasure

Global optimum



Small coins

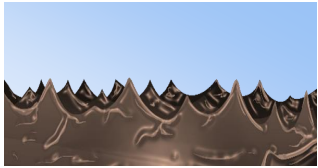
Local optima



Bob, Anthony,
Jennifer

Search agents

Analogy



Landscape



GPS: x, y

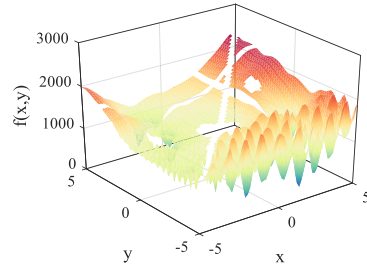


GPS: z



GPS

All the possible
locations (x, y)



Variables

Objective value

Objective function

Search space



Rain, flood, etc.

Constraints



Treasure

Global optimum



Small coins

Local optima



Bob, Anthony,
Jennifer

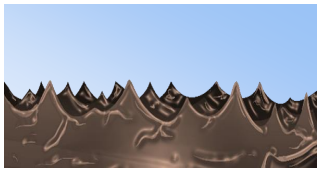
Search agents



Search strategy
(BAJ)

Optimization algorithm

Analogy



Landscape



GPS: x,y

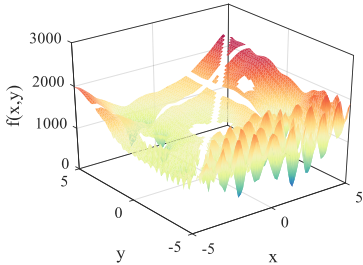


GPS: z



GPS

All the possible
locations (x,y)



Variables

Objective value

Objective function

Search space



Rain, flood, etc.

Constraints



Treasure

Global optimum



Small coins

Local optima



Bob, Anthony,
Jennifer

Search agents



Search strategy
(BAJ)

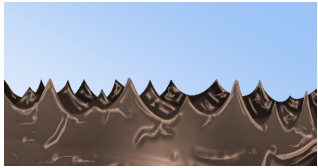
Optimization algorithm



Days

Iterations

Analogy



Landscape



GPS: x, y

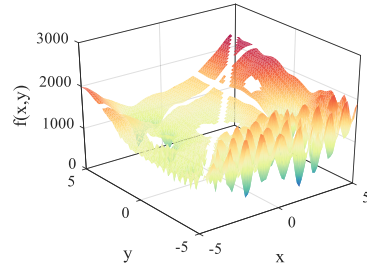


GPS: z



GPS

All the possible
locations (x, y)



Variables

Objective value

Objective function

Search space



Rain, flood, etc.

Constraints



Treasure

Global optimum



Small coins

Local optima



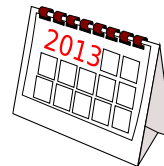
Bob, Anthony,
Jennifer

Search agents



Search strategy
(BAJ)

Optimization algorithm



Days

Iterations



The process of
finding the
best location to
find the treasure

Optimization

**Modern
Deterministic**

**vs.
vs.**

**Conventional
Stochastic**



Conventional (deterministic) optimization algorithms

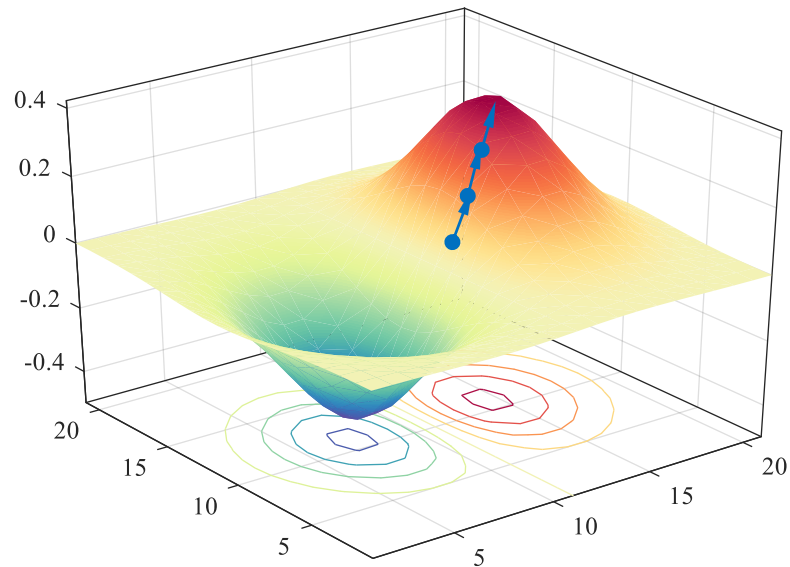


- Gradient descent algorithm

Conventional (deterministic) optimization algorithms



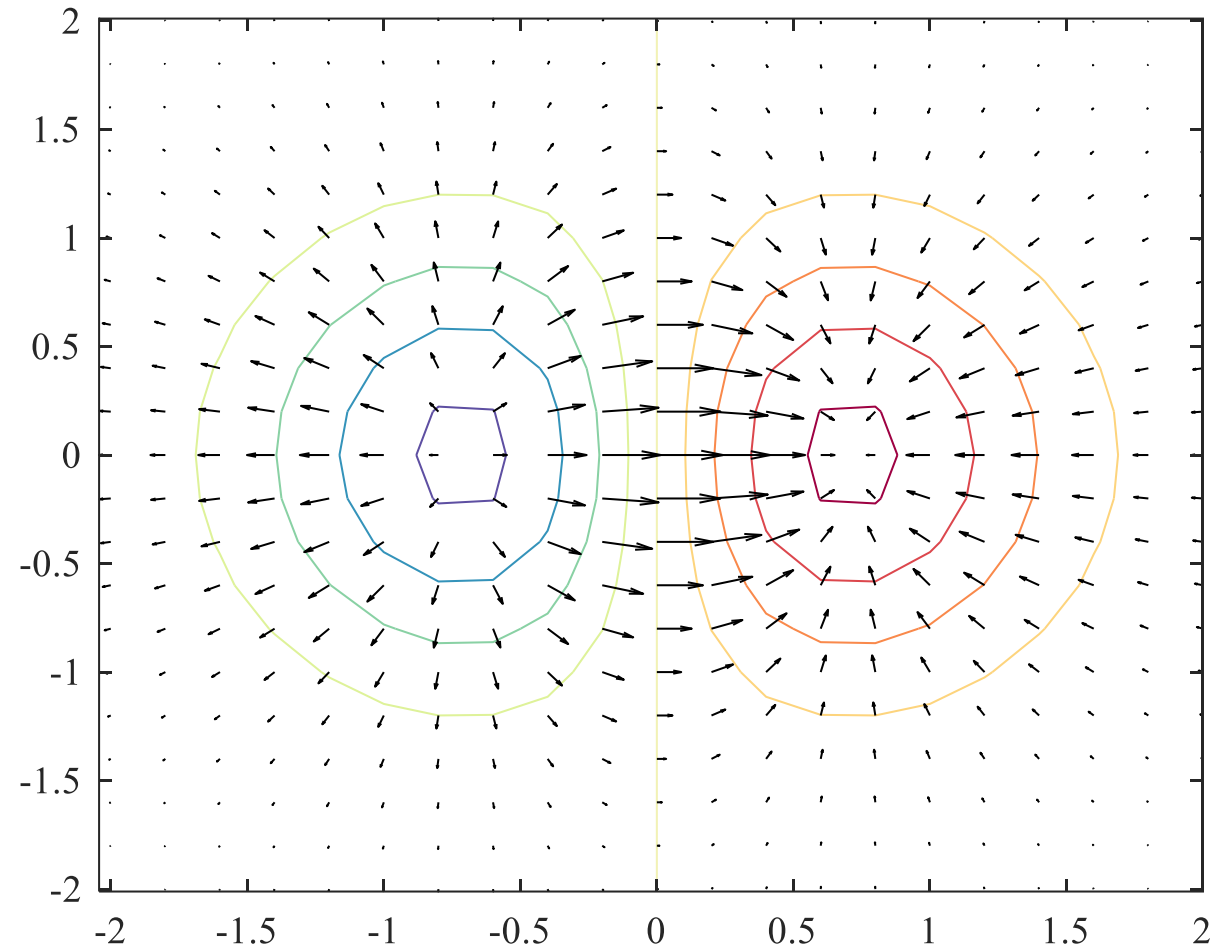
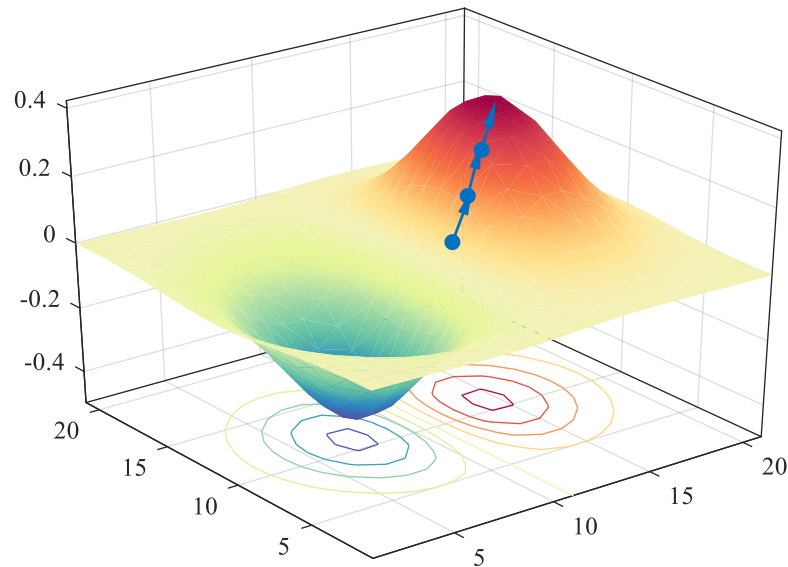
- Gradient descent algorithm



Conventional (deterministic) optimization algorithms



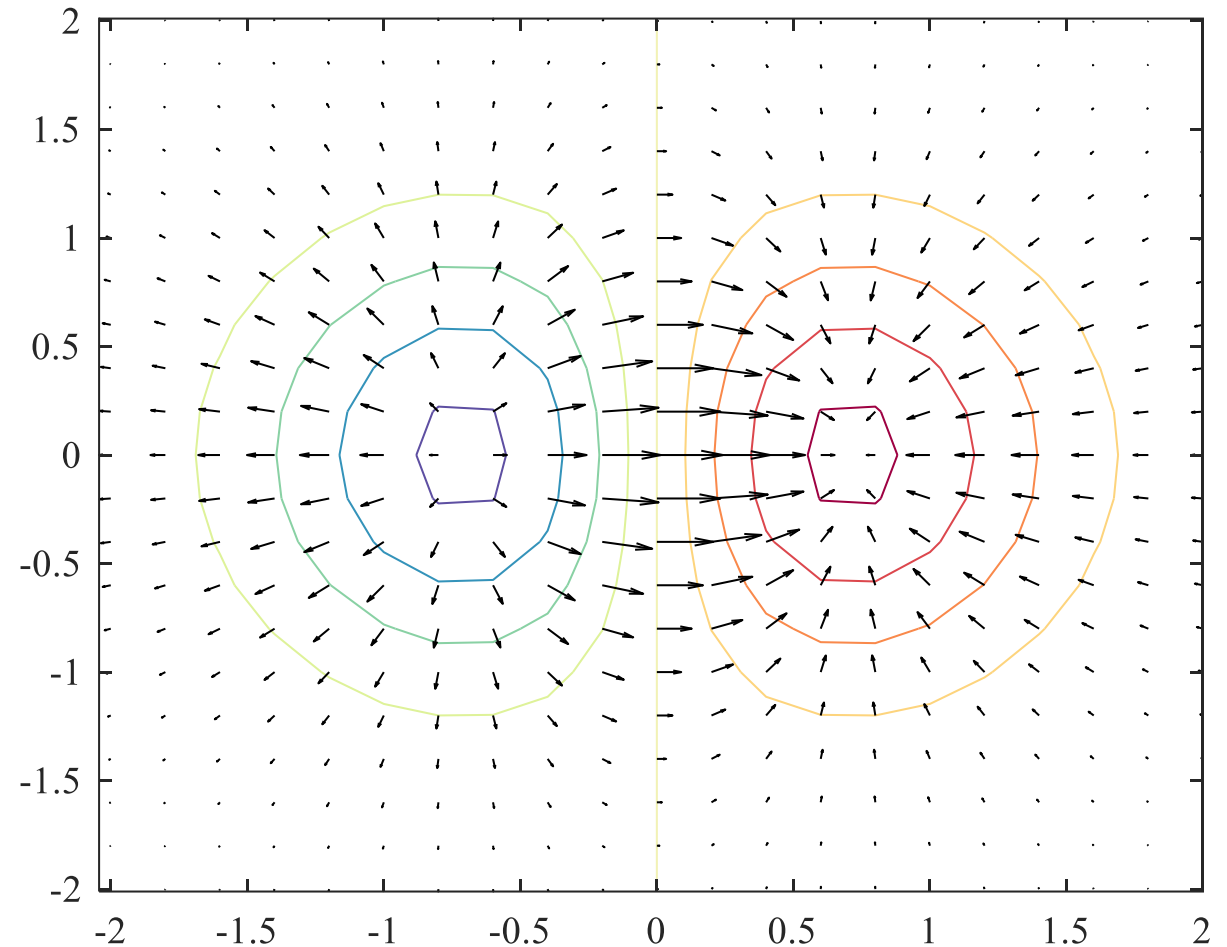
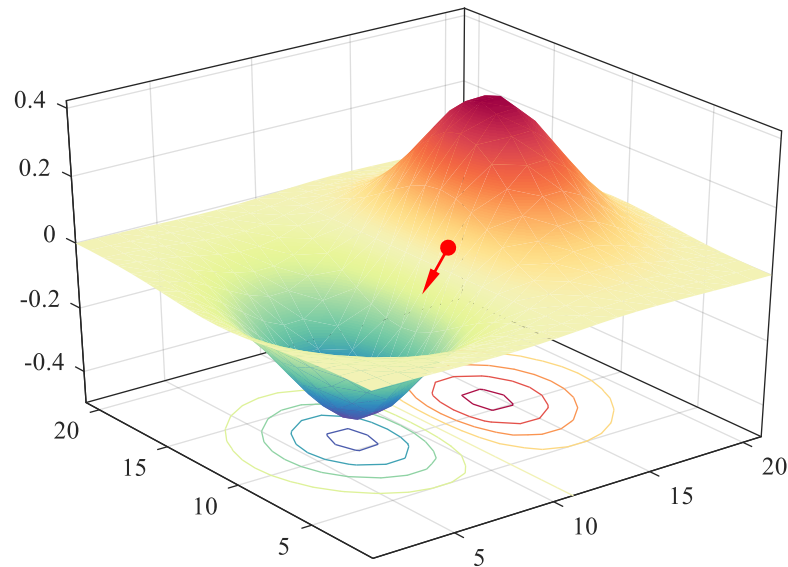
- Gradient descent algorithm



Conventional (deterministic) optimization algorithms



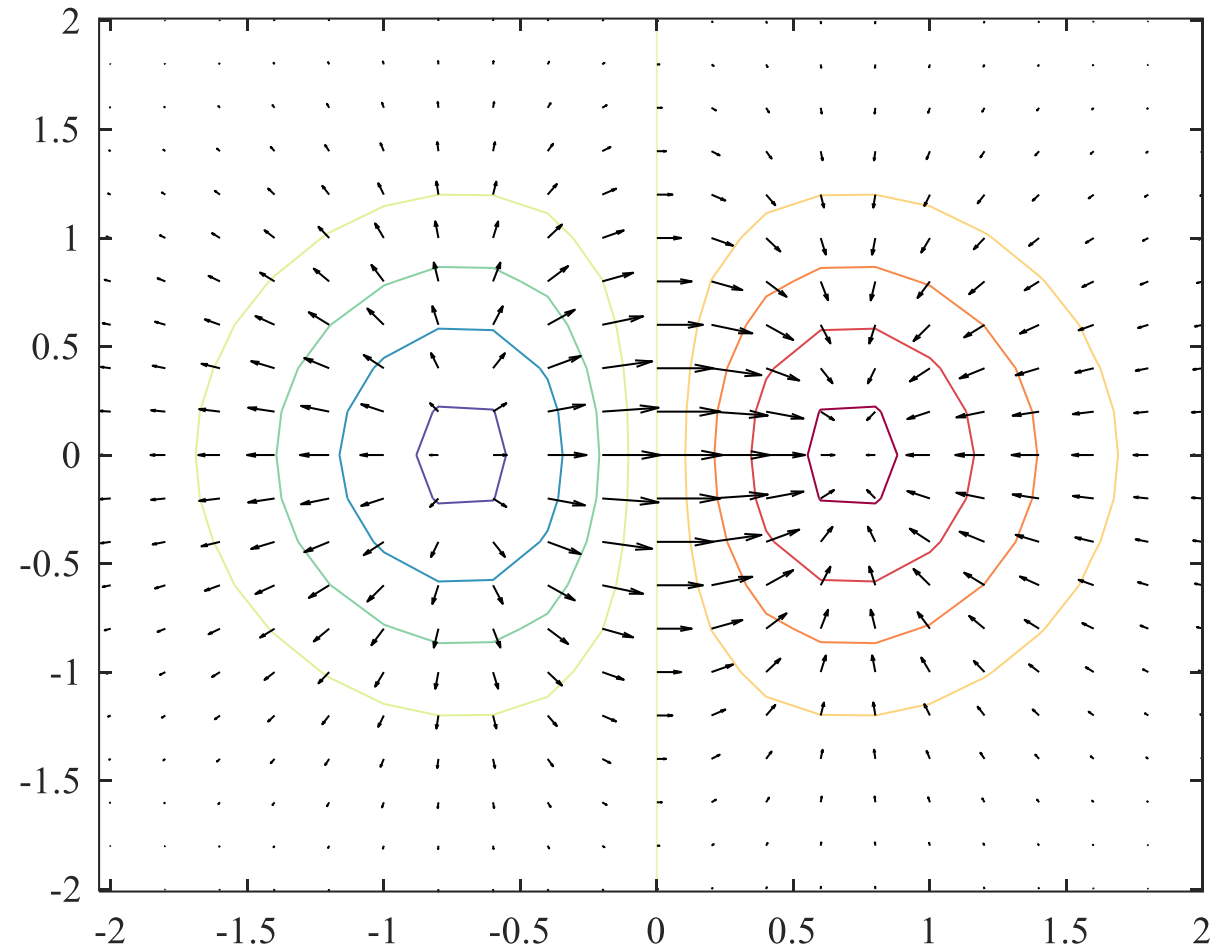
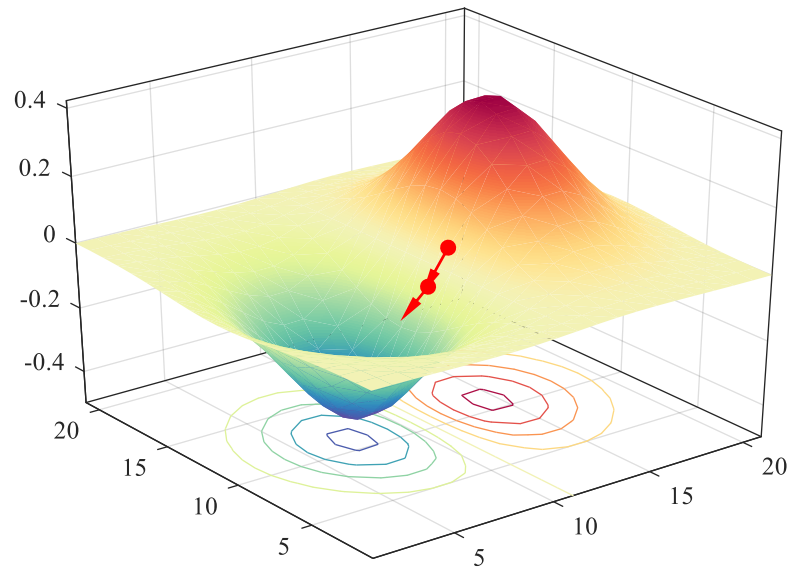
- Gradient descent algorithm



Conventional (deterministic) optimization algorithms



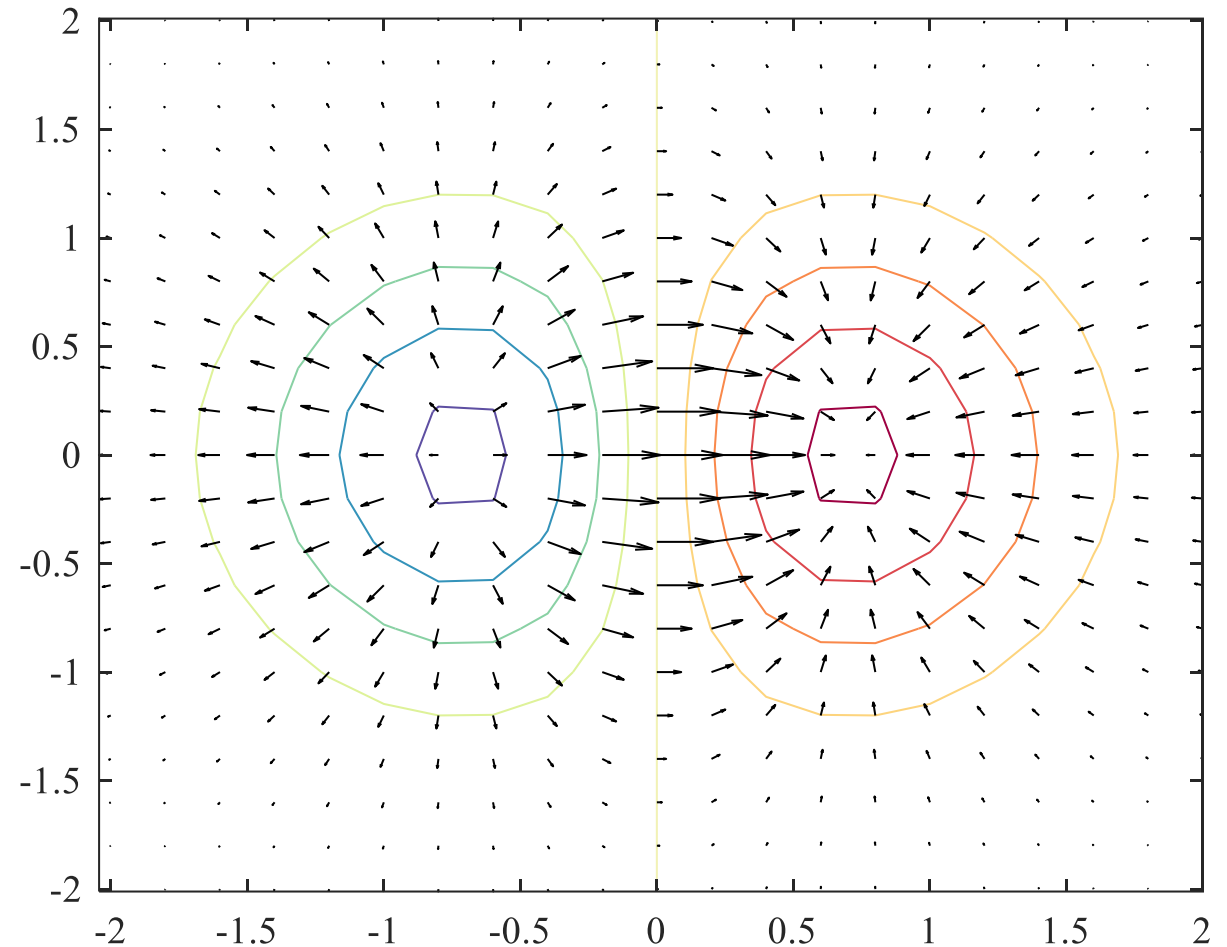
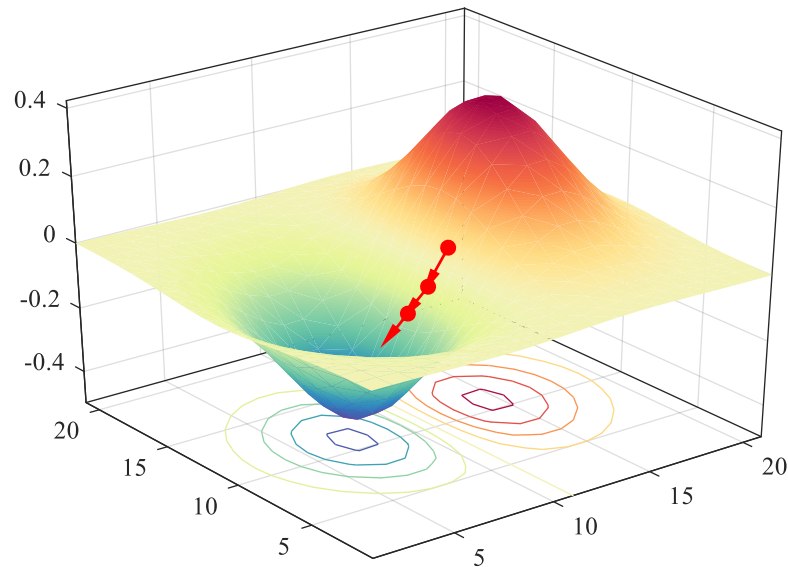
- Gradient descent algorithm



Conventional (deterministic) optimization algorithms



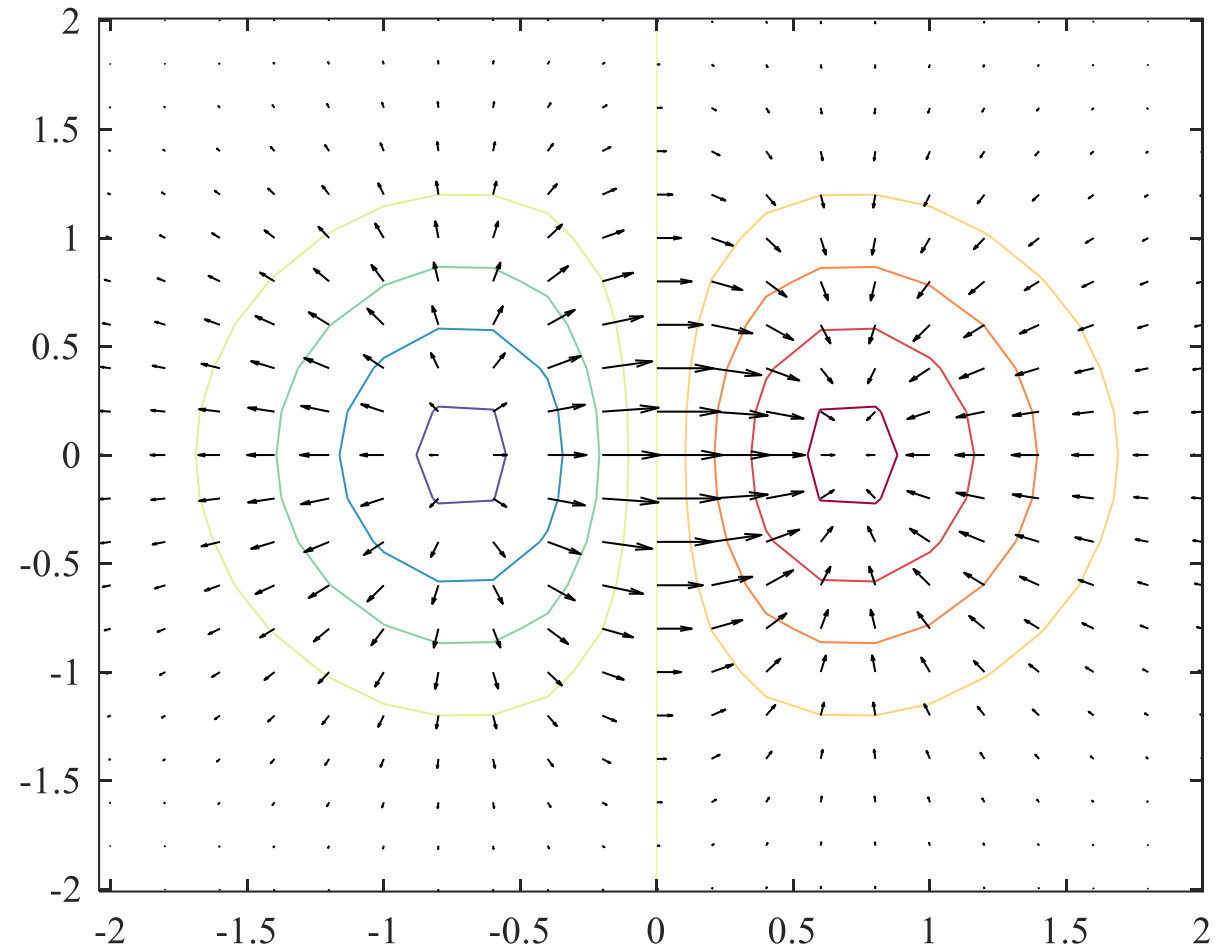
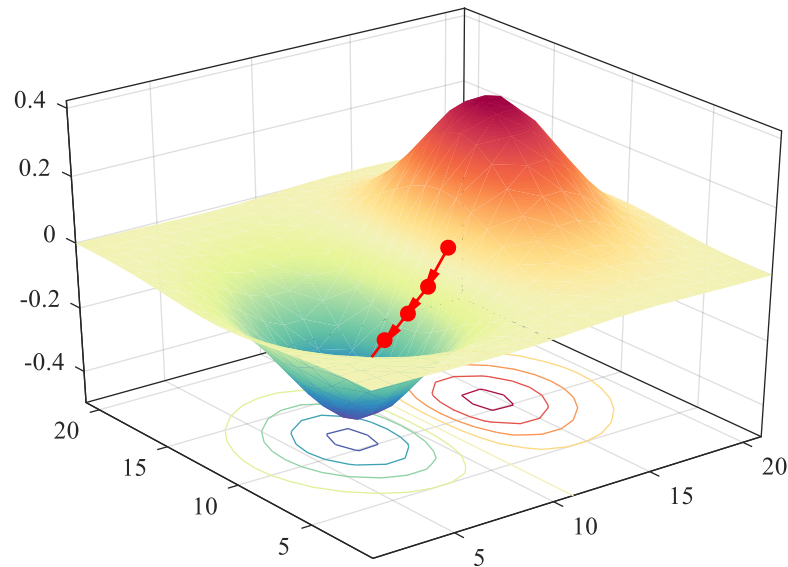
- Gradient descent algorithm



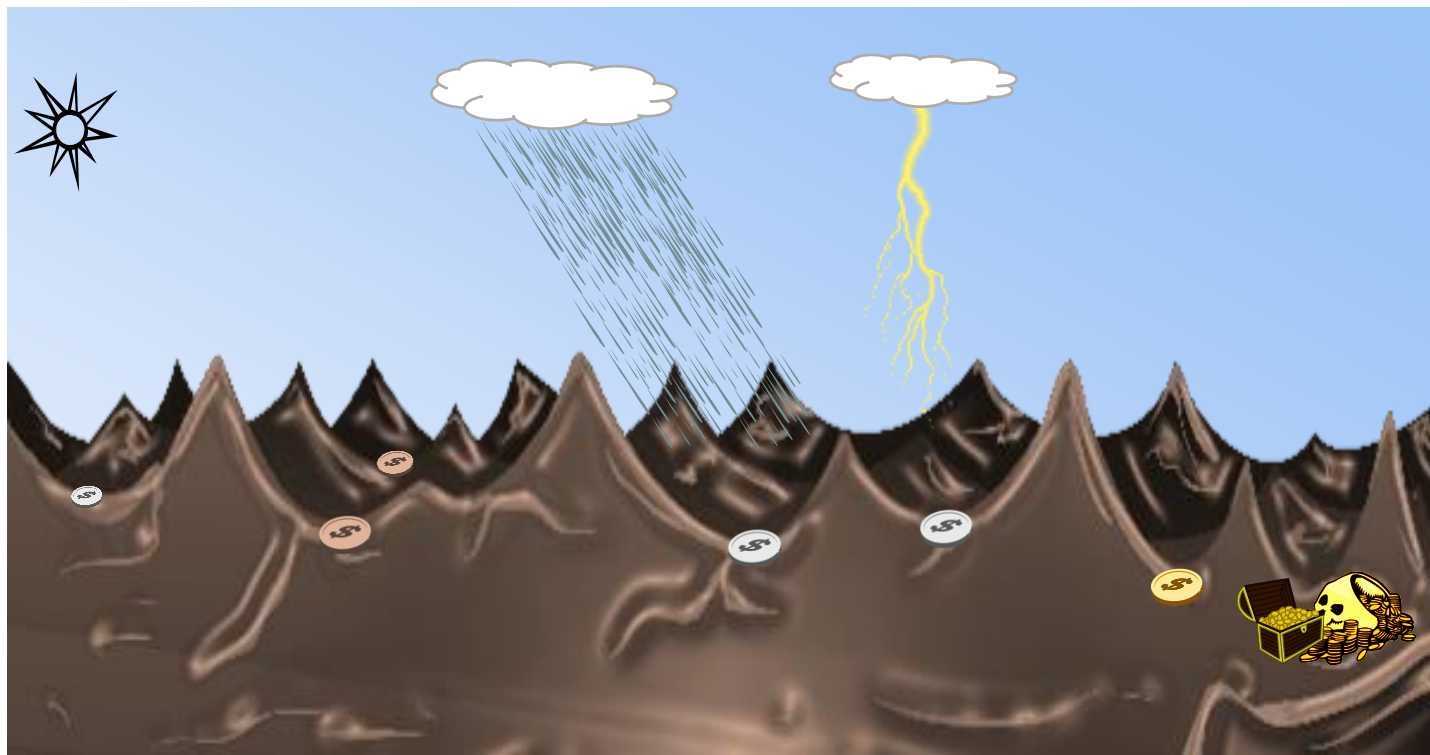
Conventional (deterministic) optimization algorithms



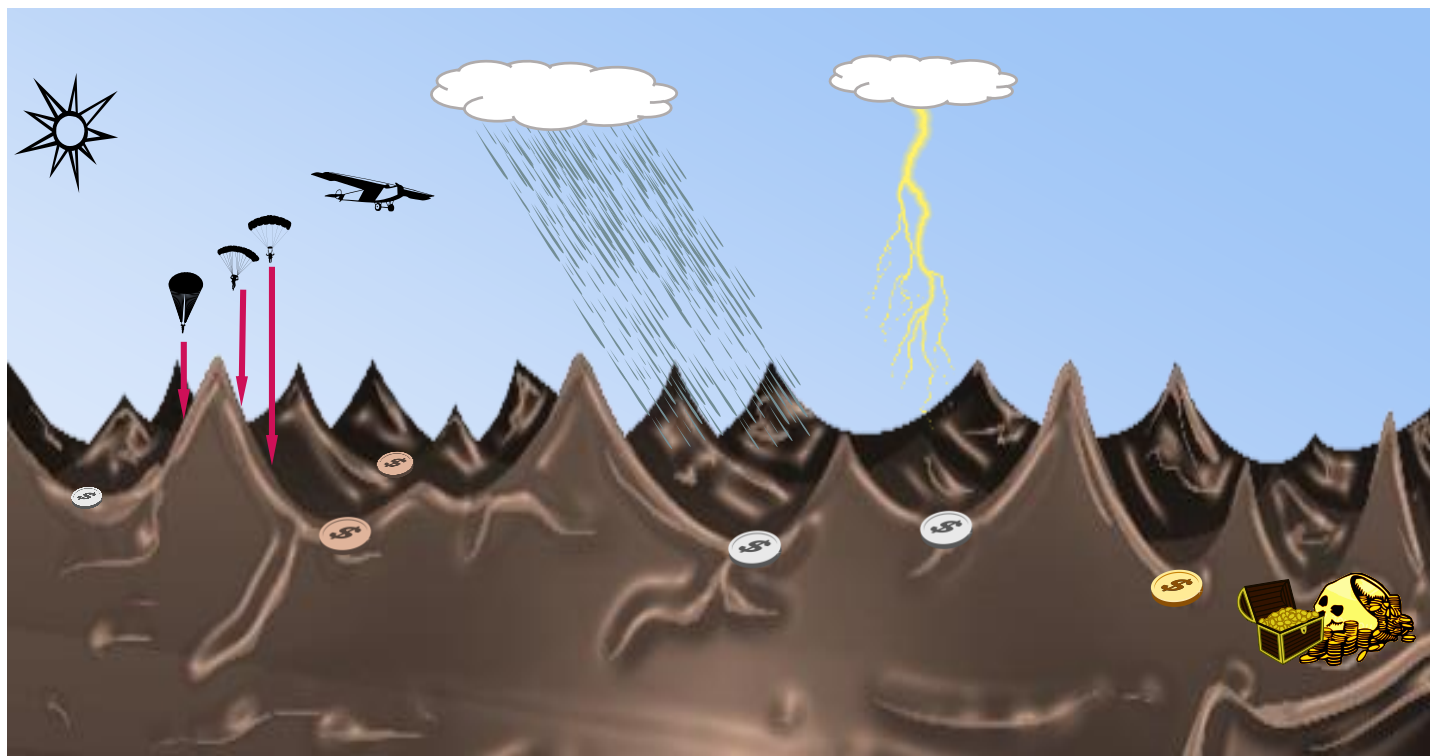
- Gradient descent algorithm



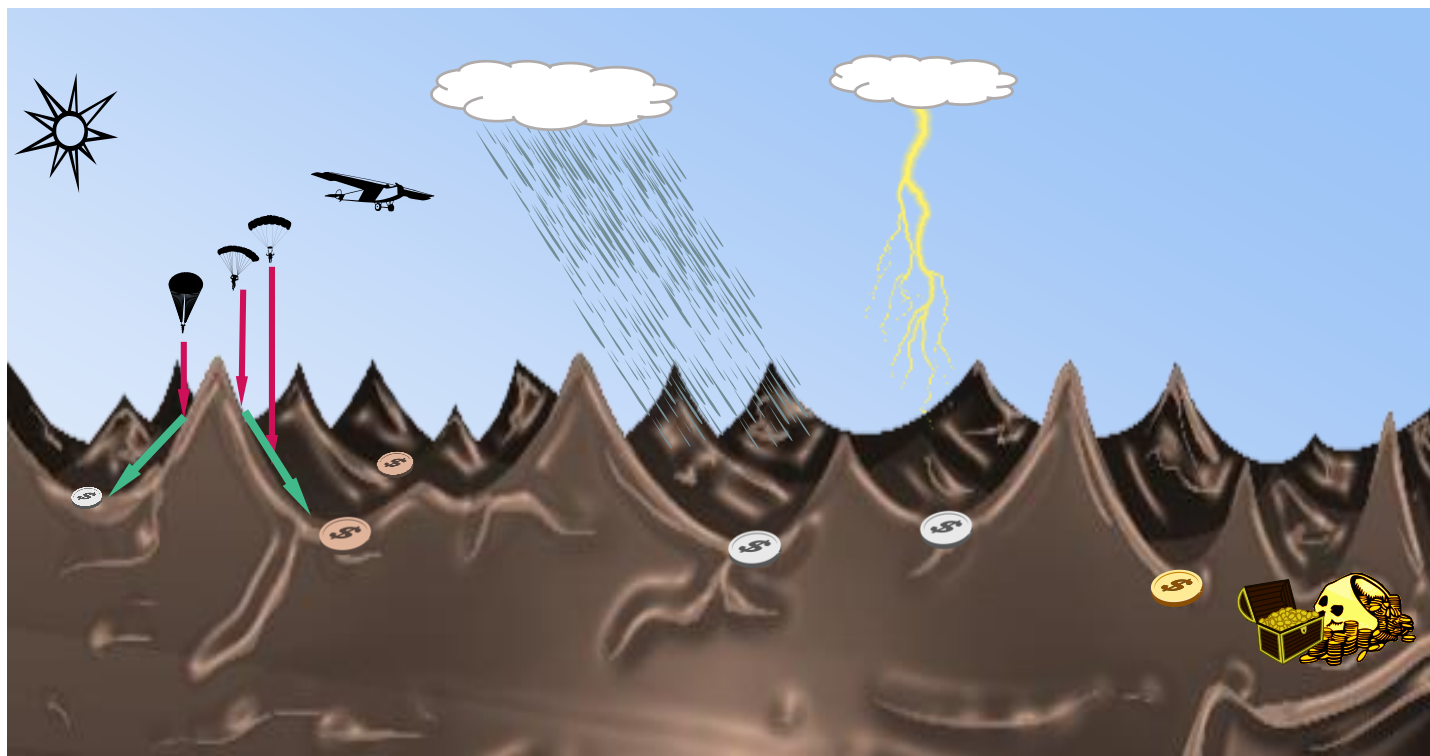
Gradient descent algorithm



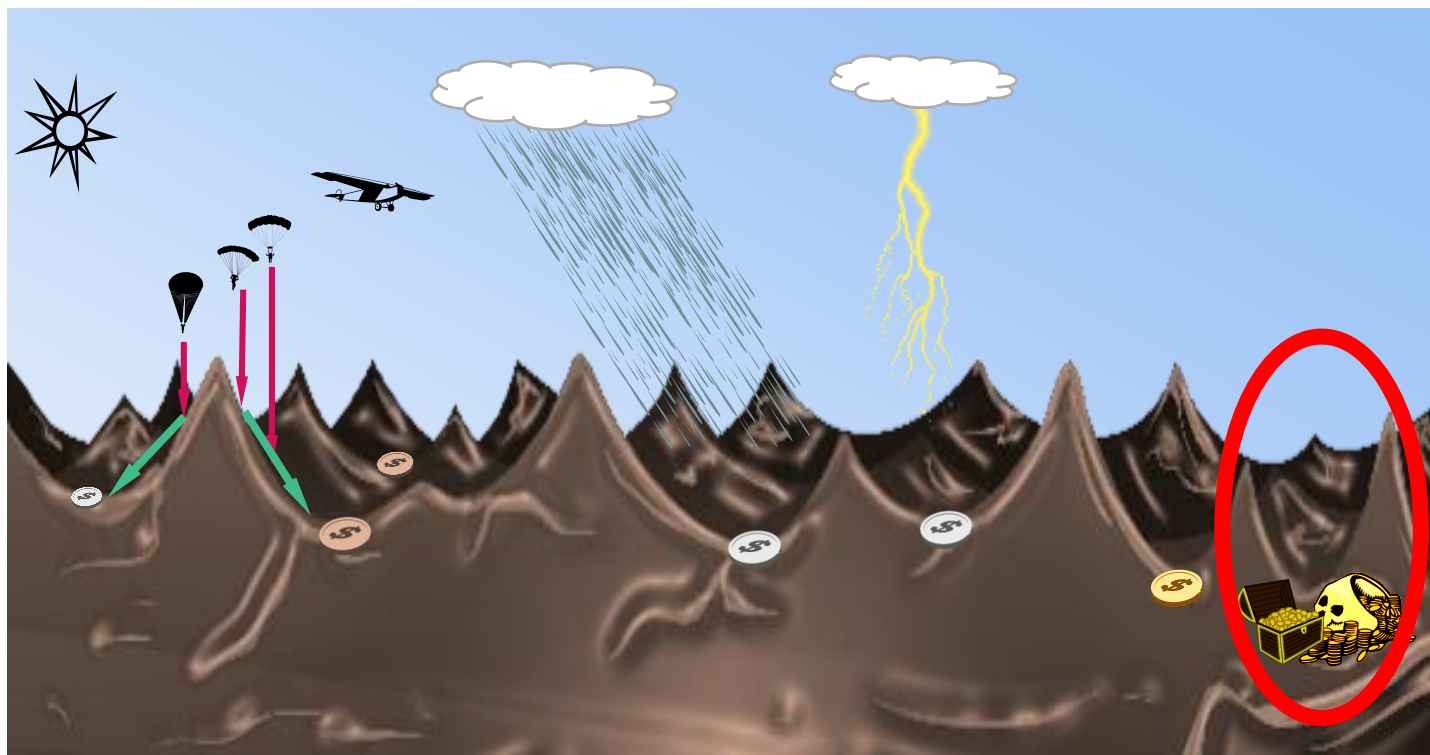
Gradient descent algorithm



Gradient descent algorithm



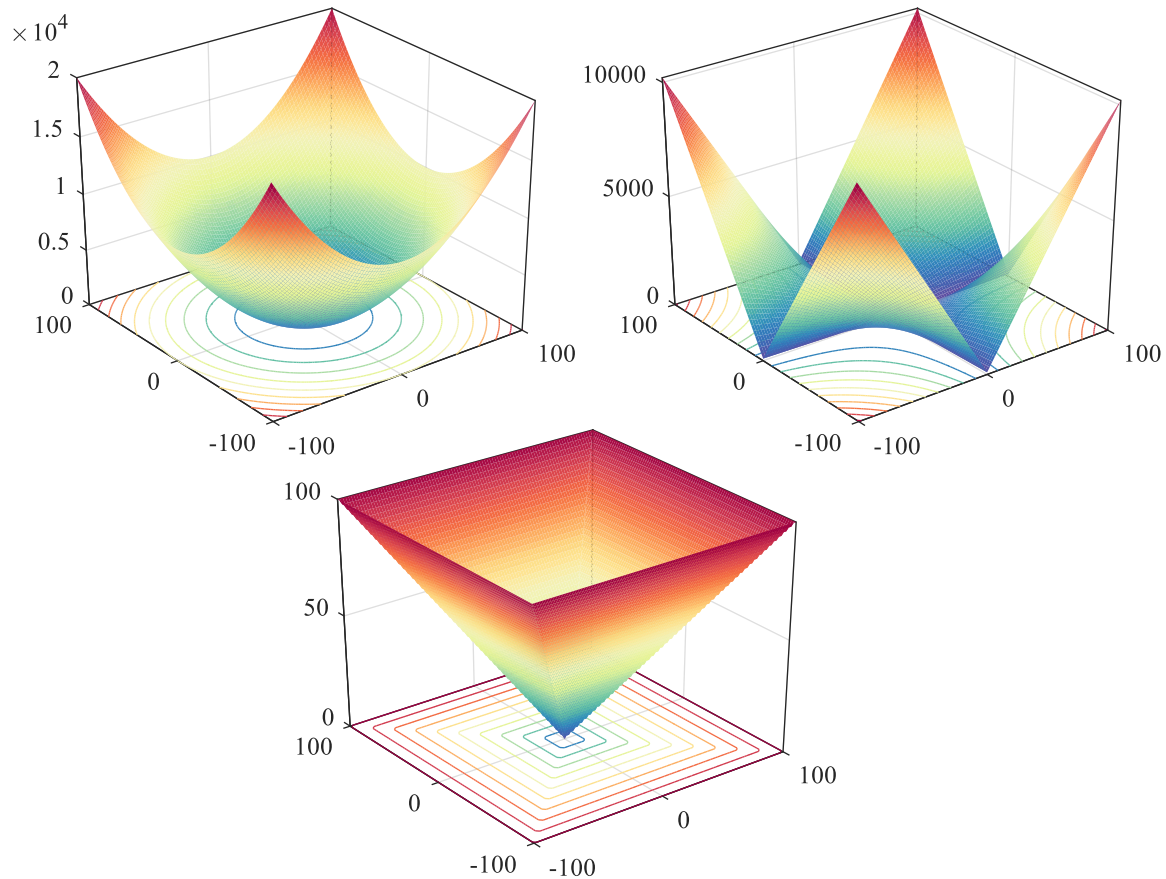
Gradient descent algorithm



Gradient descent algorithm

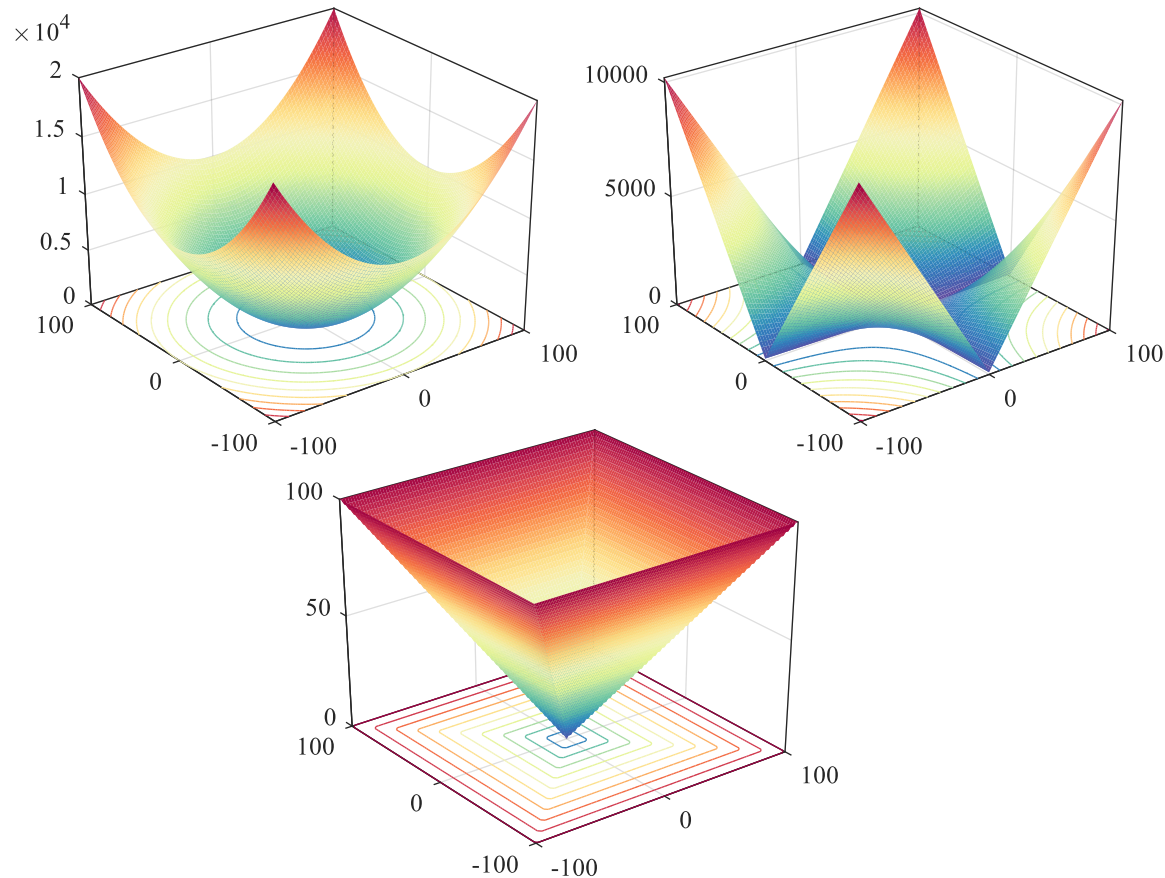
Gradient descent algorithm

Efficient for unimodal landscapes

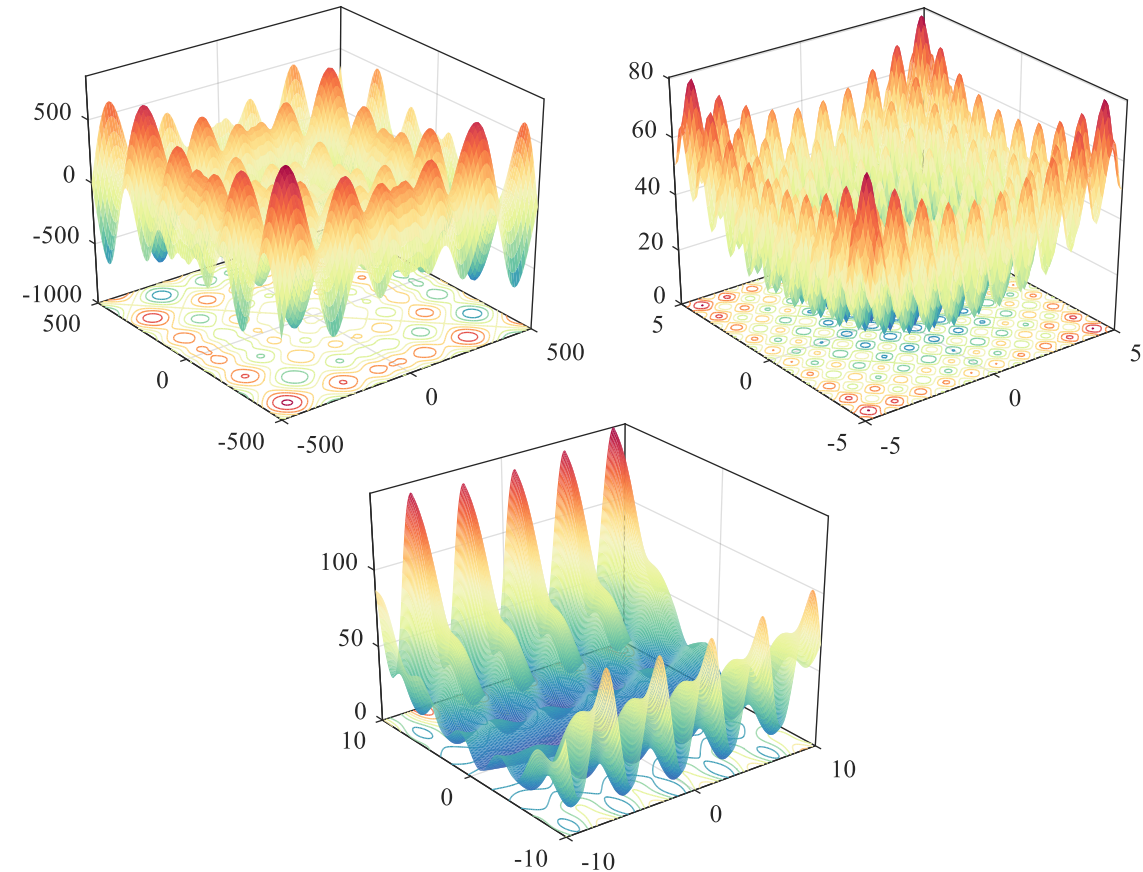


Gradient descent algorithm

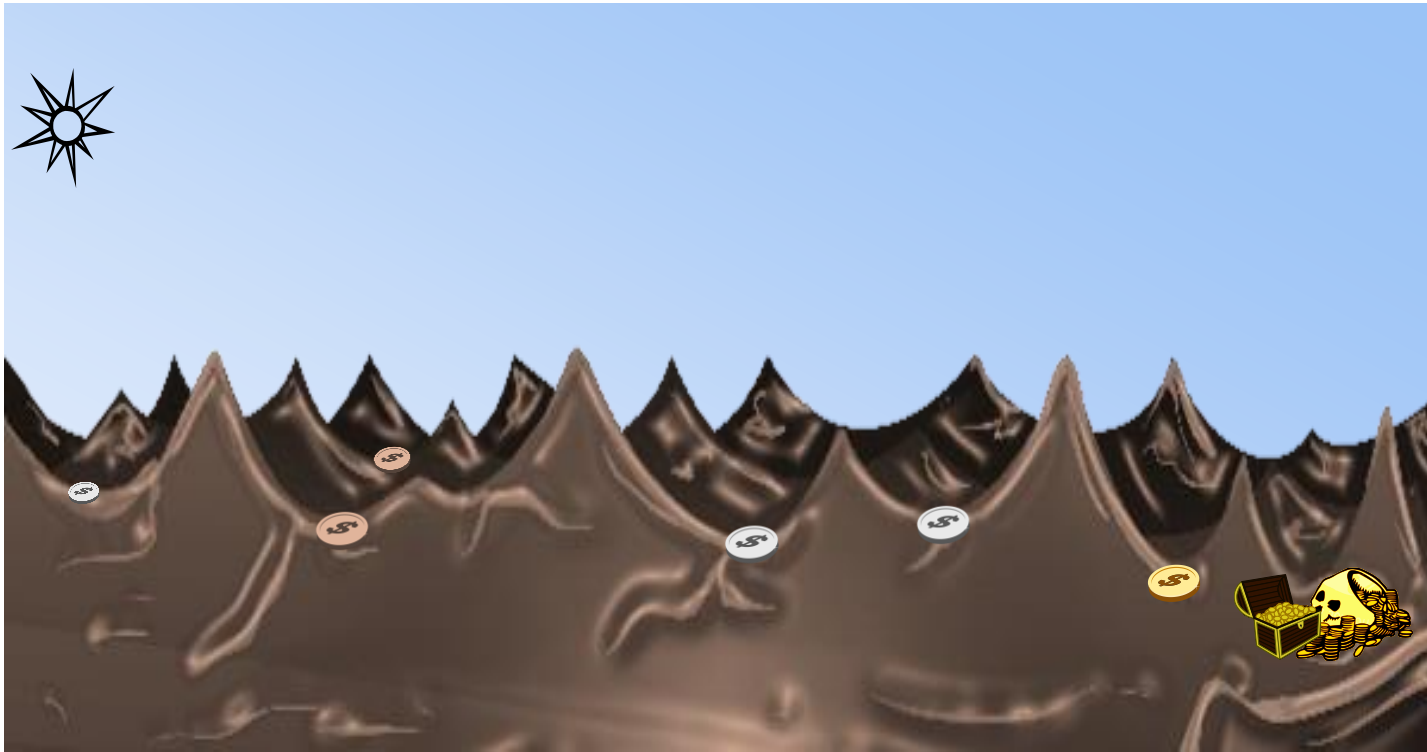
Efficient for unimodal landscapes



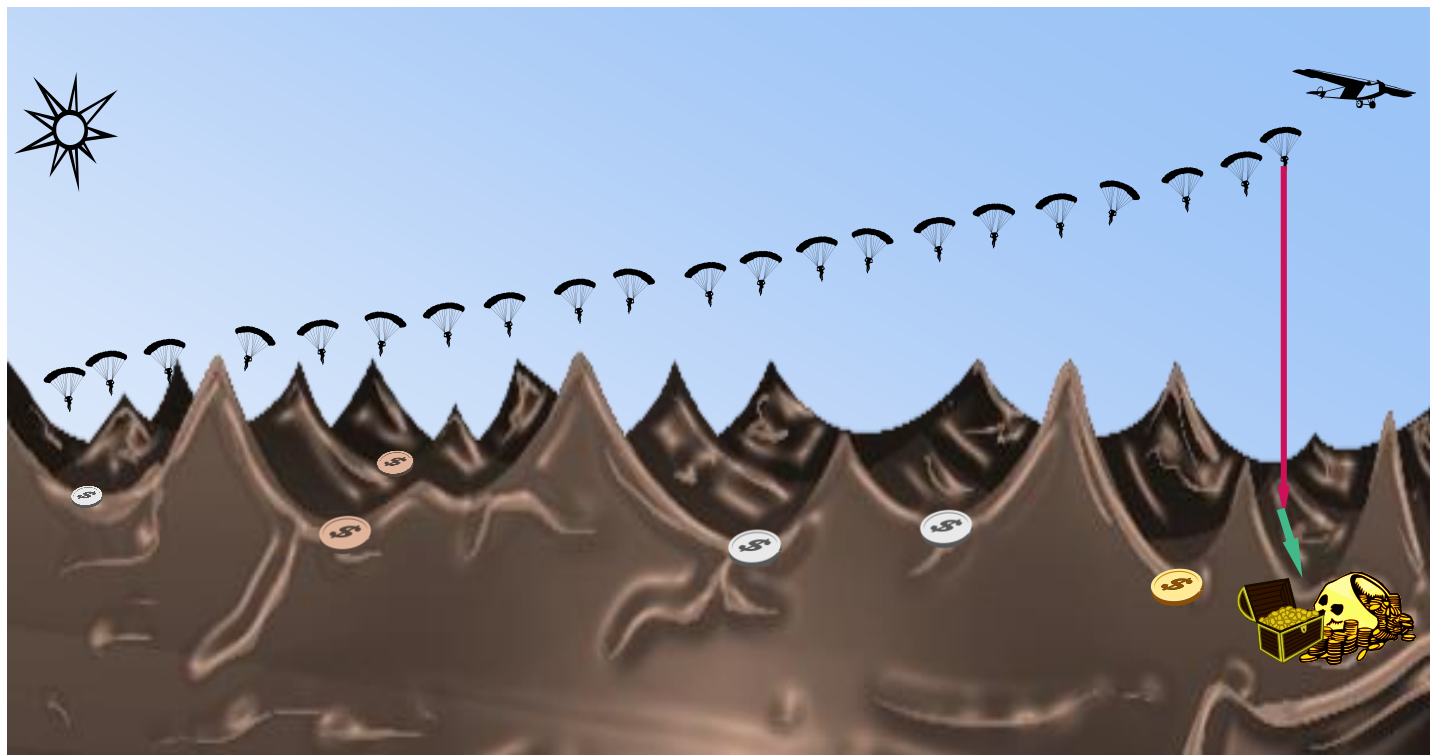
Not efficient for multimodal landscapes
Highly depends on the starting point



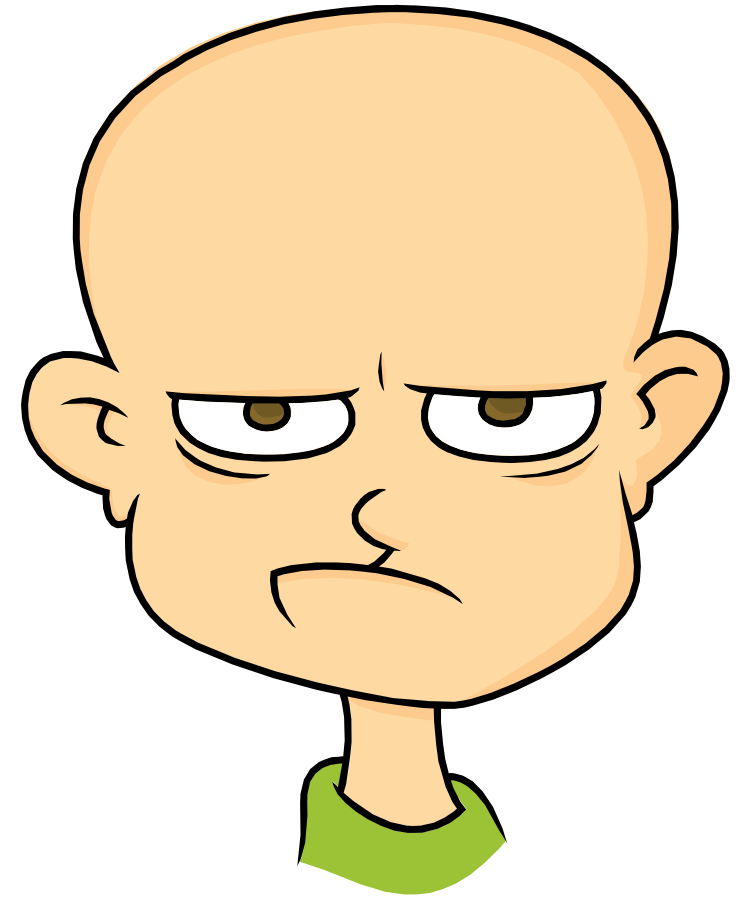
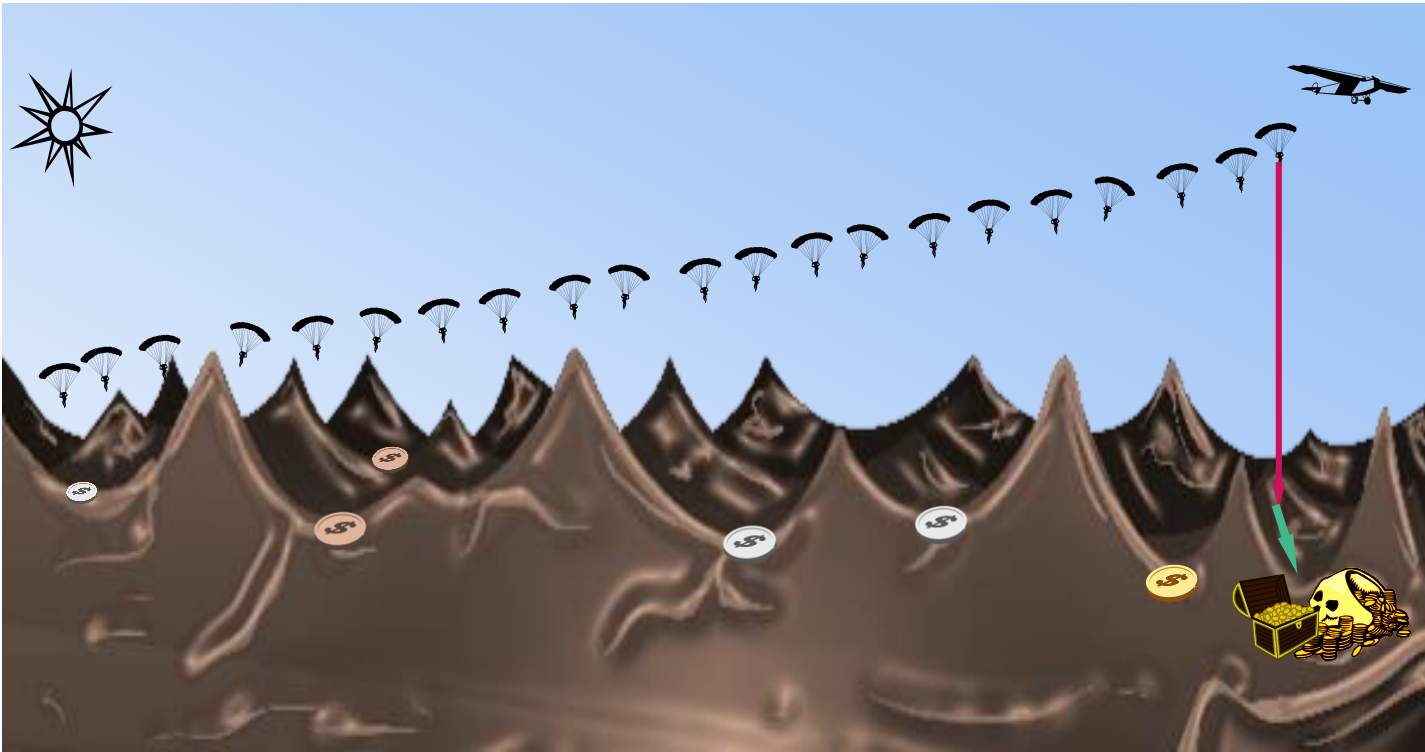
Gradient descent algorithm



Gradient descent algorithm



Gradient descent algorithm



Modern (stochastic) optimization algorithms



Deterministic algorithms

- Gradient descent algorithm

Deterministic algorithms

- Gradient descent algorithm

It gives the same output

No random component



Deterministic algorithms

- Gradient descent algorithm

It gives the same output

No random component



Stochastic algorithms

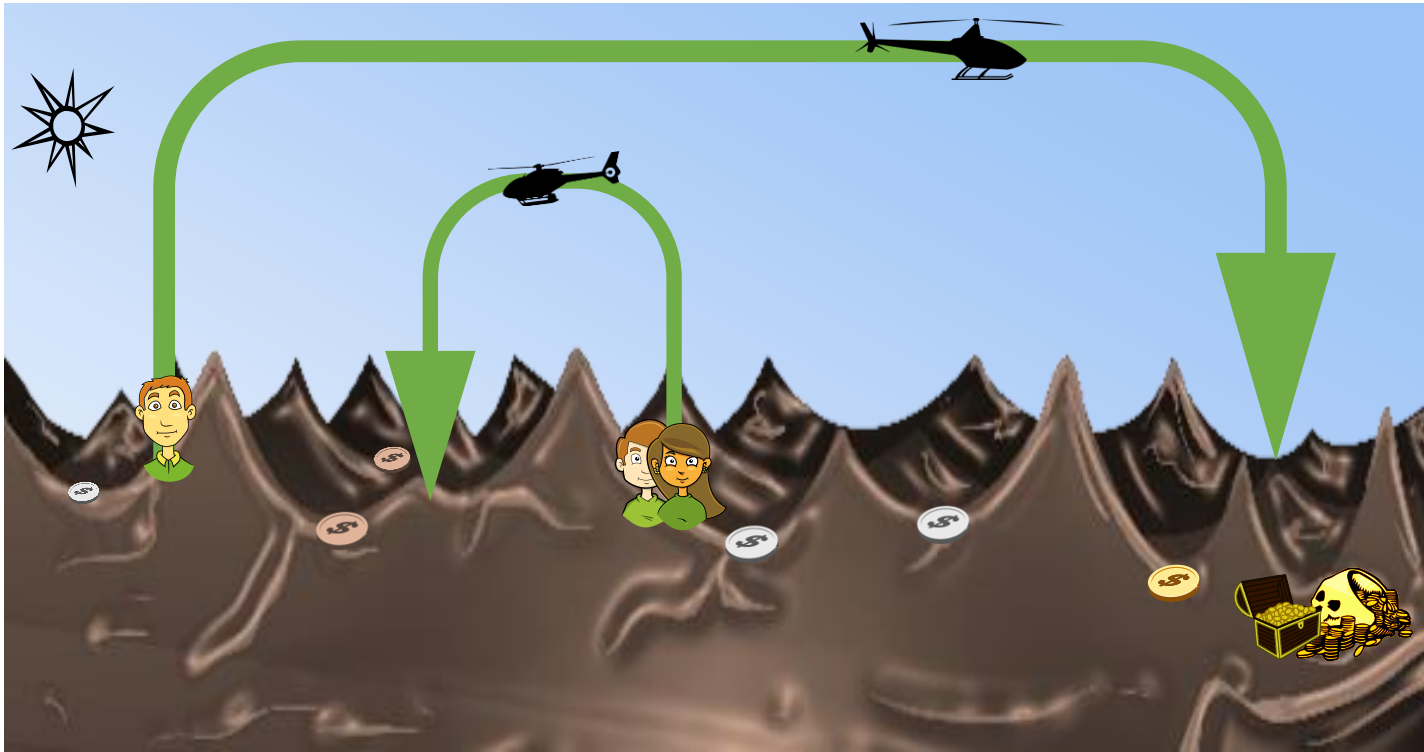
- Modern algorithms

It gives different outputs

There are random components



Modern (stochastic) optimization algorithms



Modern (stochastic) optimization algorithms

Deterministic

Advantages:

- Reliable in finding the same solution
- Require less number of function evaluation
- Fast convergence

Drawbacks:

- Local optima stagnation
- Low chance of finding the global optimum
- High dependency on the initial solution
- Mostly need gradient

Stochastics

Advantages:

- Avoid local solutions
- Higher chance of finding the global optimum
- Low dependency on the initial solution
- Mostly do not need gradient

Drawbacks:

- Slow convergence speed
- Finding different answers in each run

Modern (stochastic) optimization algorithms



High local optima
avoidance



Gradient-free
mechanism